Explainability of Text Embeddings through Semantic Perturbations and Visual Analytics

Claudia Cornacchia, Chiara Cervelli, Alessio Calcagni
Engineering in Computer Science, La Sapienza University of Rome

I. Introduction

Large Language Models (LLMs) and high-dimensional text embeddings have become the foundational infrastructure of modern Natural Language Processing (NLP). These models map text into continuous latent vector spaces, operating under the theoretical assumption that semantically similar texts remain close, while texts with divergent meanings become distant.

As embedding models grow in size and complexity, they increasingly act as "black boxes", making it difficult to understand which textual features drive their internal representations and similarity judgments. This lack of transparency is a central concern in Explainable Artificial Intelligence (XAI): complex machine learning models often achieve high predictive performance at the cost of interpretability, limiting users' ability to inspect, trust, and validate their decisions. In the context of text embeddings, this issue becomes relevant because opacity does not only affect final predictions, but also the organization of the latent space itself.

Previous literature in Explainable AI (XAI) has highlighted the risks of deploying these uninterpretable models, particularly when latent proximity is driven by superficial structural artifacts, a phenomenon often referred to as "shortcut learning" [1] or adopting "syntactic heuristics" [2], rather than actionable semantic content.

To what extent do specific syntactic structures (e.g., standard customer service phrasing) overpower core semantic meaning (e.g., operational issues) in shaping the latent space? To address this, our study examines the structural and lexical composition of input texts (independent variable) and the resulting topological shift, or "drift," of their corresponding embedding vectors (dependent variable). Our goal is to solve this explainability problem by allowing users to interactively test the model. The system targets Data Scientists and NLP Researchers as users.

The implications of these findings are highly relevant for the NLP community: they provide a necessary, human-in-the-loop diagnostic framework to identify biases and vulnerabilities, ensuring that text embeddings are robust, transparent, and reliable before real-world deployment.

II. Related Works

Our research intersects Visual Analytics for high-dimensional latent spaces, Explainable AI (XAI) in Natural Language Processing and the adversarial robustness of text embeddings. The foundational response to the opacity of high-dimensional embeddings has been the adoption of dimensionality reduction techniques, such as t-SNE [3] and [4] , to project latent spaces into interpretable 2D or 3D visual coordinates. Standard reference platforms, such as Google's Embedding Projector [5], excel at navigating these environments by allowing researchers to explore local neighborhoods, map global geometry and isolate spatial regions. However, because these traditional systems operate entirely on pre-computed data, they act as passive exploration environments. They lack the real-time, user-driven capabilities required to dynamically manipulate texts on the fly and witness how precise textual variations alter layout topology. Furthermore, while spatial projections group visually similar vectors together, they do not inherently explain why a geometric cluster was formed in the first place. Modern Visual Analytics frameworks frequently integrate topic modeling structures directly over spatial distributions. In particular, algorithms like BERTopic [6], which leverages a specialized class-based TF-IDF (c-TF-IDF) procedure, have proven highly effective at assigning abstract, human-readable labels and thematic tags to unstructured spatial clusters, turning raw geometric coordinates into readable landmarks. Recent approaches , as shown by S. Liu et al. [7] attempt to track how individual words dictate a document's 2D positioning using complex spatial word clouds or detailed mathematical matrices, which often lead to user cognitive overload. Our visual interface design instead draws inspiration from text-linked visualization tools like LSTMVis [8], which demonstrated the structural clarity of mapping hidden machine learning dynamics directly onto inline text heatmaps, an approach we leverage to communicate exact word-level anchoring weights directly onto the source text range. Lastly, a growing body of literature highlights the vulnerabilities of Large Language Models (LLMs) to adversarial text perturbations. Our work draws upon behavioral testing frameworks like CheckList [9], which advocates for testing models against conversational disfluencies, and the research of Belinkov & Bisk [10], who proved the vulnerability of neural networks to synthetic noise and natural sources of noise. Although these robustness studies are essential, they are typically conducted through offline scripts and evaluated with aggregate metrics. As a result, they offer limited insight into the contextual and case-specific topological effects that each perturbation produces within the embedding space.

III. Methodology

A. Dataset and Model Selection

For this study, we utilized the Twitter US Airline Sentiment dataset, comprising 10,445 real-world customer

service interactions. To map these texts into a high-dimensional latent space, we generated embeddings using the `qwen3-embedding:0.6B` model. We integrated the Gemma model to execute generative perturbations. We selected a generative LLM outside the Qwen family to mitigate the risk of architectural bias, ensuring that the perturbations independently challenge the embedding space rather than exploiting shared pre-training artifacts.

B. Dimensionality Reduction

To visualize the high-dimensional latent space, Uniform Manifold Approximation and Projection (UMAP) [4] was applied. High-dimensional embedding vectors were projected into a 2D space, optimizing the preservation of local topological neighborhoods while maintaining the global structure of the dataset. The UMAP parameters were configured with n_neighbors $= 15$, random_state $= 42$, min_dist $= 0.1$, and metric $=$ 'cosine' to achieve a balanced representation of the semantic clusters.

C. Topic Modeling and Keyword Extraction

To decipher the meaning of the latent clusters, the system provides dual-level explainability:

- **Global Explainability (BERTopic):** To extract human-readable latent topics across the entire dataset a BERTopic pipeline [6] (which relies on HDBSCAN for clustering) was deployed on the backend. To prevent highly frequent brand names and standard polite formulas from dominating the clusters, a custom stop-word list was applied via a `CountVectorizer` during the class-based TF-IDF (c-TF-IDF) calculation. This preprocessing step effectively isolated actionable operational keywords. During the development of the BERTopic pipeline, we encountered a fundamental Visual Analytics dilemma regarding model explainability. When computing topics on the raw dataset (*Scenario 1*, filtering only standard English stop-words), the resulting clusters were dominated by airline names and standard customer care syntax (e.g., "thanks", "thank you", "dm us"). To extract more meaningful insights about flight experiences, we applied two filters: *Scenario 2* (Removing airline names): this approach failed to surface underlying operational problems because the variance is entirely absorbed by corporate etiquette syntax. *Scenario 3* (Removing both Airline names and a list of courtesy expressions): This successfully yielded clear, actionable topics.
- **Local Explainability (TF-IDF):** For dynamic, user-selected clusters, a standard TF-IDF algorithm was implemented to extract the top 10 most distinctive keywords relative to the entire dataset, allowing users to inspect the unique lexical composition of any spatial neighborhood.

D. Word Ablation and Targeted Perturbations

To move from cluster-level analysis to instance-level interpretation, the system allows users to select a single tweet on the map and visualize it as an interactive diagnostic heatmap. To measure the semantic weight of individual words, an ablation study was conducted. For each word t in a text T, the word was iteratively removed to create a modified text T'. The impact score was calculated as the cosine distance between the embedding vectors of T and T'. The text is rendered as an inline heatmap, where the background color intensity of each word represents its semantic weight within the embedding space.

Word-Level vs. Token-Level Analysis: While the Qwen3-Embedding model processes text via sub-word tokenization, our system performs ablation and perturbations at the word level. This design choice aligns with the principle of "Interpretable Data Representations" established in XAI literature [11] , which dictates that explanations must use representations understandable to humans, regardless of the model's internal features. Furthermore, our targeted perturbations require complete words to retrieve valid lexical substitutions.

Targeted Perturbations: To test whether the model truly captures underlying semantics or merely acts as a superficial 'Bag-of-Words,' users can click individual words within the text heatmap to trigger targeted perturbations. These perturbations are structurally divided into:

1) **Word-based manipulations:** Synonym/antonym swaps (via WordNet/Datamuse integrations), Word Removal, or custom user-defined word replacements.
2) **LLM-based transformations:** Whole-tweet semantic modifications executed via the Gemma model, engineered to perform generic Paraphrasing (meaning-preserving) or Opposite generation (polarity-inverting).
3) **Noise and Structural distortions:** Automated whole-tweet attacks comprising random conversational filler injection, character-level Typos, Word Shuffling (grammatical sequence reliance tests), and Leetspeak substitution.

The Feedback Loop: Executing a perturbation triggers an instant visual feedback loop via a Kaggle backend connected to the frontend through a Ngrok tunnel. The system alters the text, extracts a new high-dimensional vector using the Qwen/Qwen3-Embedding-0.6B model, and projects it into 2D coordinates. The frontend then instantly updates the global scatterplot, drawing a visual trajectory arrow from the original "ghost" node to the new perturbed coordinate.

E. System Architecture

- **Cloud Embedding Engine:** We utilized a Kaggle Notebook equipped with an NVIDIA Tesla T4 GPU to run the `qwen3-embedding:0.6B` model. This cloud-accelerated environment ensures fast vector generation and processing.
- **API Bridge (Ngrok):** A Flask server runs within the Kaggle environment, securely exposed to the local client via an Ngrok secure tunnel, ensuring seamless communication between the cloud backend and the browser.
- **Frontend Analytics:**The interactive dashboard is built using D3.js and JavaScript for high-performance Canva rendering and real-time animation of semantic drifts.

F. Coordinated Views and Integration Design

While traditional dashboards typically implement Ben Shneiderman's classic Information Seeking Mantra (*"Overview first, zoom and filter, then details-on-demand"*) [12], our framework deals with deep machine learning black boxes.

Consequently, it embodies Daniel Keim's Visual Analytics Paradigm: "Analyse first, show the important, zoom/filter, analyse further, details on demand." [13].

The Global Map: The primary interface is an interactive 2D UMAP scatterplot where nodes represent individual tweets. To support diverse analytical tasks, the map features **dynamic color encoding**: via a dropdown menu, users can instantly recolor the entire latent space based on *Sentiment*, *Airline Company*, or *Latent Topic*. Users can pan, zoom and hover over nodes to reveal tooltips containing the raw text. When a perturbation is applied, the map renders the semantic drift in real-time: a "ghost" node remains at the original coordinate, while an arrow traces the path to the newly generated embedding coordinate. The arrow's stroke-width is dynamically scaled to encode the Euclidean distance of the drift, making massive semantic shifts immediately visually prominent.

Advanced Selection Framework: To navigate the dense UMAP projection, the system features a dual-toggle selection framework. Users can independently combine two *Interaction Modes* (Click or Brush) with two *Selection Scopes* (Point-based or Cluster-based). The dashboard dynamically adapts its panels and visual cues based on the active combination to prevent information overload.

- Point + Brush $\rightarrow$ displays subset histograms.
- Point + Click $\rightarrow$ shows single-node heatmaps.
- Cluster + Brush/Click $\rightarrow$ automatically selects macro-clusters

Lexical and semantic Panels: The system features a powerful bidirectional link between spatial selections and textual explainability.

- **Map $\rightarrow$ Panels:** Brushing a cluster on the map instantly triggers backend computations, populating two synchronized charts: a TF-IDF horizontal bar chart displaying the most distinctive exact keywords of that area, and a BERTopic distribution chart showing the detailed breakdown of abstract topics alongside clickable keywords.
- **Panels $\rightarrow$ Map:** The coordination works in reverse as well. If a user clicks a specific horizontal bar in the TF-IDF chart or a semantic tag in the BERTopic panel, the global scatterplot updates dynamically. It dims the unselected data and visually highlights only the subset of tweets containing that exact word or belonging to that specific sub-topic, allowing users to cross-reference vocabulary with spatial distribution.

IV. Results

By analyzing the high-dimensional geometry of the vector space, we isolated how the model prioritizes structural platform noise over true operational meaning.

1. Macroscopic space analysis: The failure of Sentiment and Topic clustering

An analysis of the native 1024-dimensional space revealed a striking systemic polarization. The model entirely failed to cluster texts based on emotional polarity. The global Silhouette Score for Sentiment was a negligible 0.0723. A deeper look at the distance metrics showed that the intra-cluster distance between two negative tweets (0.2314) was nearly identical to the inter-cluster distance between a negative and a positive tweet (0.2579), indicating a complete semantic overlap. Conversely, clustering by Airline Brand yielded a slightly higher Silhouette Score (0.0755), proving the vector space organizes primarily around named entities rather than emotions. A fascinating historical anomaly emerged here: the inter-cluster distance between US Airways and American Airlines was exceptionally low (0.2107). Because the dataset originates from 2015, the exact time these two airlines were merging, users frequently tagged both entities due to confusion over accountability. While this shows how the latent space captures socio-historical phenomena, it also reveals a limitation: the model lacks an external knowledge base to resolve such ambiguities, leading it to conflate distinct brands when they appear in identical contexts. When evaluating operational topics, the global Silhouette Score drops significantly (0.0101). In this unfiltered space, the operational lexicon is completely submerged by the "background noise" of generic customer care phrases (e.g., "thanks", "DM") and brand mentions, which ruin the global cluster boundaries. Thus revealing the importance of the Scenario 3 adoption.

2. Leave-One-Word-Out analysis: The dominance of Syntax over Semantics

To quantify what actually drives the model's geometry, we executed a population-scale Leave-One-Word-Out (LOWO) ablation on 10,445 tweets. The results conclusively demonstrated a structural bias toward platform-specific syntax rather than logical meaning. Removing a Twitter handle (@mention) caused an immense mean semantic drift of 0.0582, with a maximum boundary shift reaching 0.3801. This means that deleting a single @mention distorts the text's coordinates 4.13 times more than deleting a core operational term (0.0141) like "cancelled" or "baggage". Furthermore, URLs shifted the vector layout 3.33 times more than operational keywords. Strikingly, the impact was tied directly to the @ syntax rather than the brand's actual name. Standalone text brands (e.g., "united" without the handle) had a minimal impact of 0.0065, meaning the handle formatting amplifies the word's pull by nearly 9 times. Part-of-Speech (POS) metrics confirmed this superficiality: punctuation (PUNCT: 0.0333) and interjections (INTJ: 0.0366) exerted significantly more structural pull than actual nouns (0.0150) or verbs (0.0118). Crucial logical operators like negations are registered at the absolute bottom of the attention hierarchy (0.0064).

3. The Data-Driven Ablation and the "United Airlines" anomaly

We implemented a hybrid, tweet-centric ablation architecture across 93 statistically valid sub-clusters (defined as a combination of a specific Airline and Topic containing a minimum threshold of tweets). For each tweet, the system identifies both the airline handle and the most relevant operational keyword. The operational keyword is obtained through a two-tier strategy: first relying on the primary BERTopic keyword, and using a localized TF-IDF calculation (on that specific sub-cluster) as a fallback. We then created two versions of the same tweet by removing one of these anchors at a time. Both variants were passed through the Qwen3-Embedding model and compared

with the original vector. Since the two ablations come from the exact same tweet, the resulting semantic drifts (Cosine Distances) can be compared directly. To mathematically declare which word acts as the true spatial anchor, we applied the Wilcoxon signed-rank test within each sub-cluster. This non-parametric paired test evaluates whether the difference in semantic drift between the brand and the keyword is actually significant. If the p-value is strictly less than our 0.05 threshold, the algorithm confirms that the imbalance is not due to random chance, and the word that caused the highest mean semantic drift is selected as the dominant spatial anchor. Globally, the Airline Brand dominated, winning 76 out of 93 battles (81.7%). However, breaking this down by specific airlines isolated a massive statistical anomaly regarding United Airlines. While brands like Virgin America showed the highest semantic impact weighted on the number of tweets (0.099), United registered the lowest (0.032) even though it's the predominant company (13% of all tweets). Out of 19 battles involving United, the brand lost to the operational keyword 7 times and tied 6 times. From the analysis emerged that all but one of the operational keyword victories in the entire dataset belonged to United Airlines. This validates our hypothesis of "Semantic Assimilation". Because United Airlines was so overwhelmingly present as a target for negative complaints in the dataset, the model assimilated the word "United" into the baseline coordinates of a "generic airline complaint". In United's clusters, the brand name became semantically redundant, forcing the model to anchor onto concrete physical words (like baggage, wifi, or cancelled) to differentiate meaning.

4. Syntactic Vulnerability vs. Semantic Robustness (Stress-Testing)

Finally, targeted perturbation stress-tests evaluated the model's text comprehension logic. The model proved highly resilient to real-world noise, recording negligible damage from Typos (0.0127) and random Noise (0.0097). It also effectively understood latent linguistic relationships, remaining highly robust when words were replaced with Synonyms (0.0222) or Antonyms (0.0189). However, the model exhibited severe vulnerabilities when text syntax and full-context semantics were manipulated. Shuffling the word order caused a severe semantic drift (0.0586), nearly triple the impact of changing terms to synonyms. Moving beyond dictionary substitutions, executing generative contextual perturbations using an external LLM caused substantial coordinate degradation: rewriting the sentence structure via an LLM Paraphrase triggered a profound semantic drift (0.0791), while fundamentally altering the narrative meaning via an LLM Opposite impact generated an even deeper displacement (0.0933). The most catastrophic coordinate collapse occurred under the Leetspeak attack (0.1474). This hierarchical progression proves that the model successfully handles minor typographical errors and isolated vocabulary swaps, but its semantic stability fails when the entire sentence is restructured or when graphical alterations like Leetspeak are applied.

CONCLUSION

Our population-scale ablation studies, macro-geometric assessments, and targeted perturbations conclusively demonstrate that the latent space of the Qwen3-Embedding-0.6B model is heavily prioritized around surface-level structural syntax and Named Entities rather than core logico-semantic intent. By tracking the exact vector shifts across 10,445 real-world entries, our findings expose an acute vulnerability in modern dense text representations: platform-specific handles (@mentions) distort coordinate spaces over four times more than highly descriptive operational keywords (e.g., "cancelled", "lost", "baggage"). Furthermore, the model's complete failure to segregate emotional polarity, evidenced by a negligible Sentiment Silhouette Score of 0.0723 and a near-total geometric overlap between positive and negative document distributions, proves that raw vector proximity cannot be assumed to match human semantic categories.

The behavioral stress tests executed within our real-time interactive environment provide clear evidence of this architectural limitation. While the embedding space exhibits strong micro-level resilience to minor typos and vocabulary swaps, it displays extreme structural fragility to structural rephrasing and graphical mutations. Furthermore, standard conversational courtesy markers and brand proper nouns act as dominant semantic shields, flattening underlying distribution variance and making it difficult to extract actionable insights.

Ultimately, these insights demonstrate the risks of applying dimensionality reduction or clustering pipelines directly to raw text streams without structural mediation. Without advanced filtering, the resulting spatial layouts form misleading topologies. Our visual analytics framework resolves this interpretability bottleneck by offering an interactive, human-in-the-loop tool.

By combining visual exploration with real-time text perturbations, we enable NLP researchers and data scientists to instantly see what textual features truly drive the model's behavior. This framework provides a foundation that future studies could use to benchmark multiple embedding architectures. By scaling up the interactive perturbation strategies, following research could evaluate how different models react to text alterations and determine whether these syntactic biases disappear in larger models.

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